

Taicheng Guo

Updated June 27, 2026

Email: tguo2@nd.edu

[Homepage](#)

[Google Scholar](#)

[Github](#)

RESEARCH INTERESTS

My long-term research & engineering goal is to build systems for AGI. My research centers on **Agents** and **Systems**, and how they co-evolve on the path to superintelligence, especially Effective Long-Horizon LLM Agent Training for AI-System / Complex-Task Automation (Recursive Self-Improvement, Code Intelligence, AutoResearch), Agent-System Co-Design (Inference Optimization), and Language Model Post-Training. I work as a research engineer who is driven to build novel & fundamental systems from scratch.

EDUCATION

University of Notre Dame

(Jan. 2023 - July 2026 (Flexible)) US

Ph.D. in Computer Science and Engineering

Advisor: Prof. Xiangliang Zhang. GPA: 4.0/4.0

Thesis: From Answering to Autonomous Research: Language Models for Research-Level Tasks

Committee: Drs. Xiangliang Zhang, Nitesh V. Chawla, Meng Jiang, Wei Wang (UCLA), Chandan K. Reddy (Virginia Tech).

King Abdullah University of Science and Technology (KAUST)

Saudi Arabia

M.S. in Computer Science

Advisor: Prof. Xiangliang Zhang. GPA: 3.89/4.0

Xidian University

Bachelor in Software Engineering. GPA: 3.5/4.0

WORK&INTERN EXPERIENCE

ND-IBM (Research Assistant)

Jun. 2026 – present

Topic: Training LLM Agents with Self-Improving Verifiers for Fine-Grained Credit Assignment.

Amazon (Applied Scientist Intern)

Jan. 2025 – Jul. 2025

Topic: Training Long-Horizon Coding Agents.

[\[ACL 2026 Main\]](#), [\[Received Full-Time Return L5 Incline Offer\]](#)

Amazon (Applied Scientist Intern)

Jun. 2024 – Sep. 2024

Topic: Building Long-Horizon Agents within Database for Personalization. [\[KDD@RelSciFM\]](#)

MBZUAI (Research Intern)

Aug. 2022 – Oct. 2022

Topic: Training Conditional-Based Diffusion Model. [\[ICASSP 2024\]](#)

KAUST (Visiting Student)

Dec. 2020 – Aug. 2021

Topic: Training Language Models for Cross-Lingual News Recommendation. [\[WWW 2023 Oral\]](#)

Tuyoo Games (Startup), AI Platform (Senior Research Engineer)

Aug. 2018 – Oct. 2020

Topic: Led a 4-person team on Game-AI intelligence, building zero-sum game agents via Monte Carlo Tree Search (MCTS), Minimax, etc.

[\[Employee of the Year Award \(Top 2%\), Employee of the Quarter Award twice \(Top 2%\)\]](#)

Qunar Topic: Strategy Product Manager (Build Personalization Product) Jul. 2017 – May. 2018

ACADEMIC COMPETITIONS

KDD Cup Large-Scale RecSys Competition. (2nd / 1895 teams, \$4,000) [\[Link\]](#)

NeurIPS Black-Box AutoML Optimization Challenge. (2nd / 200+ teams, W.S. track) [\[Link\]](#)

Kaggle Text Sentiment Analysis. (5th / 80+ teams) [\[Link\]](#)

IEEE Computer Society Data Science challenge. (1st place, \$1000) [\[Link\]](#)

SELECTED PUBLICATIONS

▸ Citations: 3110 (80%+ from first-author papers)

Full publication can be found at: [link](#)

[1] [\[Long-Horizon Agent Training, AI-System Automation\]](#) AutoLLMResearch: Training Research Agents for Automating LLM Experiment Configuration – Learning from Cheap, Optimizing Expensive

Taicheng Guo, Nitesh V. Chawla, Olaf Wiest, Xiangliang Zhang

In Submission

- [2] [\[Agent-System Co-Design, AI-System Automation\]](#) **ASAP: Agent-System Co-Design for WallTime-Centered Auto HPO Research for ML Experiments**
Taicheng Guo, Haomin Zhuang, Kehan Guo, Nitesh V. Chawla, Olaf Wiest, Xiangliang Zhang
In Submission
- [3] [\[Long-Horizon Agent Training, AI-System Automation\]](#) **MTSQL-R1: Towards Long-Horizon Multi-Turn Text-to-SQL via Agentic Training**
Taicheng Guo, Hai Wang, ChaoChun Liu, Mohsen Golalikhani, Xin Chen, Xiangliang Zhang, Chandan K. Reddy
In ACL 2026 Main, Acceptance rate: 19%, 40+ Github Stars
- [4] [\[Agent-System Co-Design\]](#) **Large Language Model based Multi-Agents: A Survey of Progress and Challenges**
Taicheng Guo, Xiuying Chen, Yaqi Wang, Ruidi Chang, Shichao Pei, Nitesh V. Chawla, Olaf Wiest, Xiangliang Zhang
In IJCAI 2024, Acceptance rate: 20%, Top-1 Most Influential Paper, 1k+ Github Stars
- [5] [\[Language Model Post-Training\]](#) **ReactionTeam: Teaming Experts for Divergent Thinking Beyond Typical Reaction Patterns.**
Taicheng Guo, Changsheng Ma, Xiuying Chen, Bozhao Nan, Kehan Guo, Shichao Pei, Nitesh V. Chawla, Olaf Wiest, Xiangliang Zhang
In IEEE Conference on BigData 2025 Oral, Acceptance rate: 18.7%
- [6] [\[LLM Reasoning\]](#) **What can Large Language Models do in chemistry? A comprehensive benchmark on eight tasks**
Taicheng Guo, Kehan Guo, Bozhao Nan, Zhenwen Liang, Zhichun Guo, Nitesh V. Chawla, Olaf Wiest, Xiangliang Zhang
In NeurIPS 2023, Acceptance rate: 32.7%, Top 2% Influential Paper, 150+ Github Stars
- [7] [\[Language Model Post-Training\]](#) **Few-shot News Recommendation via Cross-lingual Transfer**
Taicheng Guo, Lu Yu, Basem Shihada, Xiangliang Zhang
In WWW 2023 Oral, Acceptance rate: 19.2%
- [8] [\[LLM RL Training, Reward Hacking\]](#) **Domain-Adaptive Reasoning model for Injury Classification from Incident Descriptions**
Taicheng Guo, Hai Wang, Shucheng (Sue) Zhuang, Mohsen Golalikhani, Chaochun Liu, Wei Wang, Xin Chen
In AMLC 2025 RL workshop, Cross-team collaboration work
- [9] [\[Agent-System Co-Design\]](#) **Knowledge Graph Enhanced Language Agents for Recommendation**
Taicheng Guo, Chaochun Liu, Hai Wang, Varun Mannam, Fang Wang, Xin Chen, Xiangliang Zhang, Chandan K. Reddy
In KDD@RelSciFM Workshop Oral
- [10] [\[AI-System Automation\]](#) **Data Interpreter: An LLM Agent For Data Science**
Sirui Hong, ... Taicheng Guo as a contributor to Graph-based Agent Planning.
In ACL Findings 2025, 68k+ Github Stars